

1. Setting

Let $A \subseteq \mathbb{R}^{d_A}$ be the arm set and let $D \subseteq \mathbb{R}^{d_D}$ be the outcome set. We assume that A and D are Euclidean balls of radii R_A and R_D (not necessarily centred at the origin). An element $x \in A$ is called an **arm**, while an element $y \in D$ is a possible sampled outcome. The reward is affine in the arm and outcome:

$$r(x, y) := a^T x + b^T y + c.$$

Let $Z \subseteq \mathbb{R}^{d_Z}$ denote the hypothesis set, and let $z^* \in Z$ be the unknown true hypothesis. For each arm $x \in A$, define the true feasible outcome set

$$K^*(x) := \{y \in D : F_0(x, z^*) + F_1(y, z^*) = 0\},$$

where F_0 and F_1 are both bilinear maps.

After fixing z^* , the constraint can be written in intrinsic affine coordinates as

$$Cy + Bx + d = 0,$$

where B , C , and d are fixed once z^* is fixed. Equivalently,

$$K^*(x) = \{y \in D : Cy + Bx + d = 0\}. \quad (1)$$

We assume throughout that $K^*(x)$ is nonempty for every $x \in A$.

The interaction proceeds for rounds $t = 1, \dots, T$. Let $\mathcal{F}_{t-1}$ denote the history before round t . The learner chooses an arm $x_t \in A$. Nature may then choose, adaptively as a function of the history and x_t , any distribution supported on D whose conditional mean

$$m_t := E[y_t \mid \mathcal{F}_{t-1}, x_t]$$

belongs to $K^*(x_t)$. An outcome y_t is sampled from this distribution, and the learner receives reward $r(x_t, y_t)$.

The robust value of an arm and the optimal robust value are

$$v^*(x) := \min_{y \in K^*(x)} r(x, y), \quad V^* := \max_{x \in A} v^*(x).$$

The regret over horizon T is

$$R_T := TV^* - \sum_{t=1}^T r(x_t, y_t).$$

Finally, let

$$C_r := \max_{x \in A, y \in D} r(x, y) - \min_{x \in A, y \in D} r(x, y)$$

denote the reward range.

Theorem 1 (efficient $T^{\frac{8}{9}}$ learning). *Suppose that the known problem data in the setting above are rational. For every known horizon T , there is a policy whose arithmetic and weak-optimization running time is polynomial in T and the input bit length and which, for every true hypothesis and every compatible adaptive nature policy, satisfies*

$$E[R_T] \leq \tilde{O}\left(P(d_A, d_D, R_A, R_D, \|b\|_2, C_r)T^{\frac{8}{9}}\right),$$

where P is a fixed polynomial. No geometric property of Z is used beyond its inducing a valid family of compatible sets of the form Equation 1.

2. Warmup: Noiseless Setting

In this section, playing an arm x reveals an exact feasible response $y \in K^*(x)$; there is no sampling noise. However, there still is the Knightian uncertainty corresponding to the adversary picking an arbitrary outcome inside $K^*(x)$.

Lemma 1. *There exist an affine map $f : A \rightarrow \mathbb{R}^{d_D}$ and a fixed linear subspace $N \subseteq \mathbb{R}^{d_D}$ such that*

$$K^*(x) = (f(x) + N) \cap D, \quad x \in A.$$

Equivalently, before intersecting with D , the feasible sets form a family of parallel affine spaces.

Lemma 2. *Suppose that, for every arm $x \in A$, we have identified a nonempty set $K'(x) \subseteq K^*(x)$, and suppose that whenever the learner plays x , nature chooses an outcome in $K'(x)$. Define*

$$v'(x) := \min_{y \in K'(x)} r(x, y),$$

and choose

$$x' \in \operatorname{argmax}_{x \in A} v'(x).$$

Then playing x' on every round incurs nonpositive regret with respect to the original robust benchmark, i.e. $R_T \leq 0$.

Lemma 3. *There exist $d_A + 1$ arms $x^0, \dots, x^{d_A} \in A$ such that, from any possible sequence of responses $y^i \in K^*(x^i)$, we can compute an affine map $\hat{f}$ such that*

$$K^*(x) = (\hat{f}(x) + N) \cap D, \quad x \in A.$$

Proof. Let c_A be the centre of A , let $e_1, \dots, e_{d_A}$ be the standard basis vectors, and choose

$$x^0 := c_A, \quad x^i := c_A + R_A e_i, \quad i = 1, \dots, d_A.$$

These points form an affine basis. Suppose their observed responses are $y^0, \dots, y^{d_A}$, and let $\hat{f}$ be the unique affine map satisfying

$$\hat{f}(x^i) = y^i, \quad i = 0, \dots, d_A.$$

Thus $\hat{f}$ can be computed by affine interpolation from the observed responses.

Fix an affine map f and a linear subspace N as in Lemma 1, and let π be the quotient map modulo N . Since $y^i \in K^*(x^i) \subseteq f(x^i) + N$, we have

$$\pi(y^i) = \pi(f(x^i)), \quad i = 0, \dots, d_A.$$

Applying π to either $\hat{f}$ or f gives an affine map into the quotient space. These two quotient-valued affine maps agree on the affine basis $x^0, \dots, x^{d_A}$, so they agree everywhere. Consequently,

$$\pi(\hat{f}(x)) = \pi(f(x)), \quad x \in A.$$

Equivalently, $\hat{f}(x) - f(x) \in N$, and hence

$$\hat{f}(x) + N = f(x) + N.$$

Using Lemma 1 and intersecting with D therefore gives

$$K^*(x) = (\hat{f}(x) + N) \cap D, \quad x \in A.$$

This proves the claim. ■